

## MODULE 18

# Mockito for Test Isolation

Learn how to use Mockito to create mocks, isolate dependencies, and test business logic with confidence.







# MODULE 18 OVERVIEW

*Learn how to use Mockito to create mocks, isolate dependencies, and test business logic in Java applications.*

- Real code depends on databases, external services, and other classes. Mockito lets you replace those dependencies with controllable test doubles, so you can test business logic in complete isolation.

## DURATION & LAB



### 1 Hour Theory

Mockito architecture, mocking, stubbing, verification.



### 2 Hours Lab

Mockito and Mocking.



### Scenario

Test DefaultCustomerService in isolation by mocking CustomerRepository.

## FOCUS

Test Isolation, Reliability, Maintainability

## CORE SKILLS

Create mocks, stub methods, verify interactions

## THE PAYOFF

Fast, deterministic, dependency-free unit tests

**KEY TAKEAWAY** Mockito lets you test business logic in isolation — no database, no external services, just your code under test.



# WHY MOCKING MATTERS

*Testing against real dependencies makes tests slow, flaky, and hard to control.*

## WITHOUT MOCKING

- Tests hit a real database or external service
- Slow, flaky, and order-dependent tests
- Hard to simulate error scenarios
- Failures may be caused by infrastructure, not code

VS

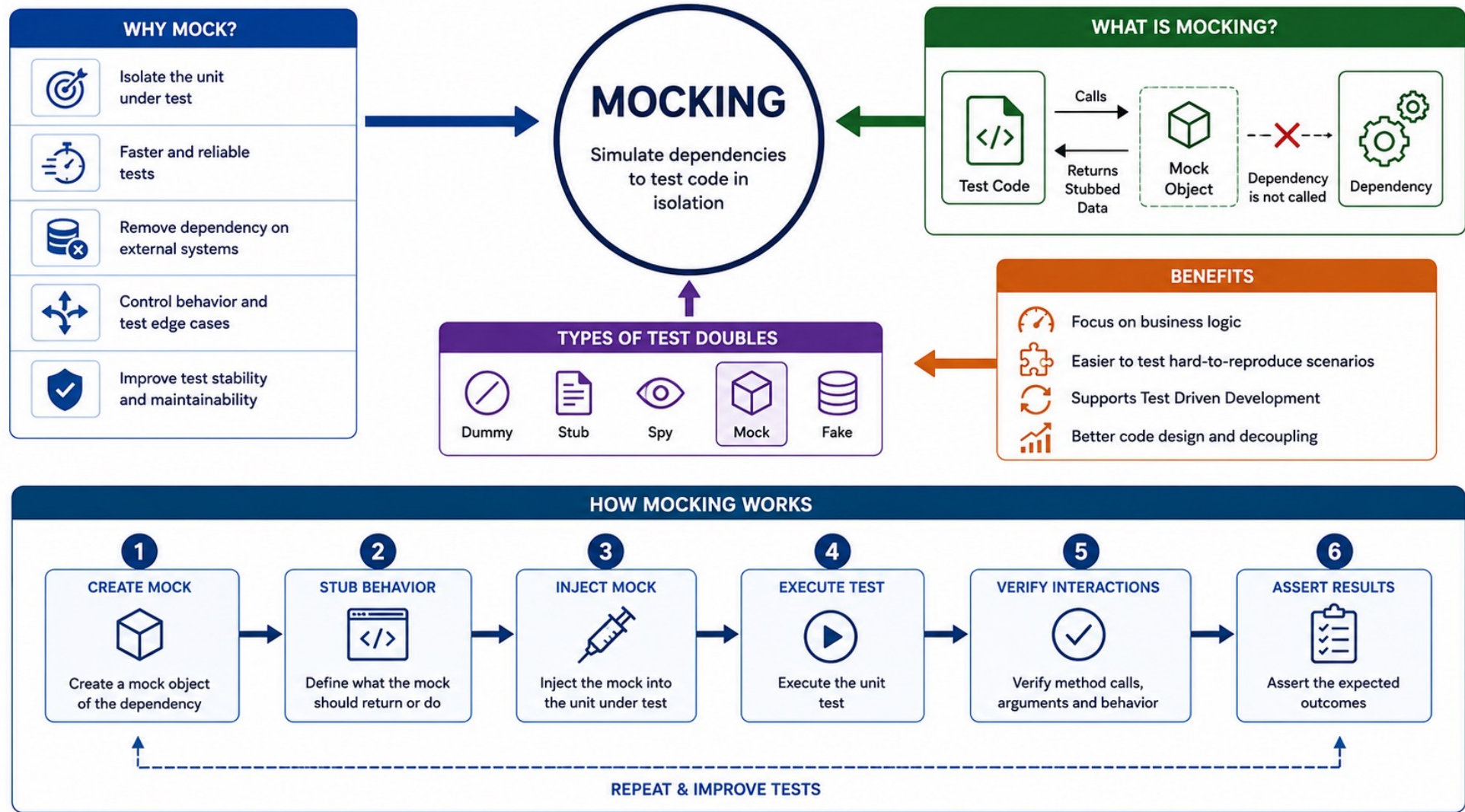
## WITH MOCKING

- Dependencies replaced with controllable test doubles
- Fast, deterministic, isolated tests
- Easy to simulate success and failure scenarios
- Failures point directly to the unit under test

## MOCKING ALLOWS US TO

- Speed up tests — no real calls to databases, APIs, or external systems.
- Improve reliability — tests don't fail due to external issues outside your control.
- Gain control over dependencies — define exactly what a dependency returns or does.
- Test edge cases easily — simulate exceptions, timeouts, and unusual scenarios.
- Encourage clean, loosely coupled code that's more maintainable and testable.

**KEY TAKEAWAY** Mocking isolates the unit under test so failures point to real bugs, not flaky infrastructure.



# TEST DOUBLES OVERVIEW

*"Test double" is the umbrella term for objects that stand in for real dependencies during testing.*

| TYPE  | PURPOSE                                      | EXAMPLE                                               |
|-------|----------------------------------------------|-------------------------------------------------------|
| Dummy | Passed but never actually used               | A placeholder object required by a method signature   |
| Stub  | Returns predefined answers to calls          | <code>when(repo.findById(1L)).thenReturn(user)</code> |
| Mock  | Stub + records interactions for verification | <code>verify(repo).save(any(User.class))</code>       |
| Spy   | Wraps a real object, allows partial stubbing | <code>spy(new OrderService())</code>                  |
| Fake  | A working, simplified implementation         | An in-memory repository instead of a DB               |

**Purpose:** replace real collaborators with controlled objects so you can test one unit of code in isolation.

**Note:** These categories describe roles, not necessarily different libraries or object types. A Mockito mock may be used only as a stub in one test. A fake is usually a working implementation, not created by Mockito. A spy wraps or partially delegates to a real object. A mock is most useful when interaction verification matters.

**Takeaway:** Choose the simplest test double that gives the test the control it needs. Do not default to a mock when a real value object or small fake is clearer.

**KEY TAKEAWAY** Mockito creates configurable test doubles that can be used for stubbing, interaction verification, or partial delegation.

# MOCKS VS. SPIES

*A mock starts empty and returns defaults; a spy wraps a real object so only the methods you stub change.*

## EXAMPLE

```
List<String> mockList = mock(List.class);
when(mockList.size())
    .thenReturn(42);

List<String> real = new ArrayList<>();
List<String> spyList = spy(real);
spyList.add("mockito");

doReturn(100)
    .when(spyList).size();

verify(spyList).add("mockito");
```

## KEY POINTS

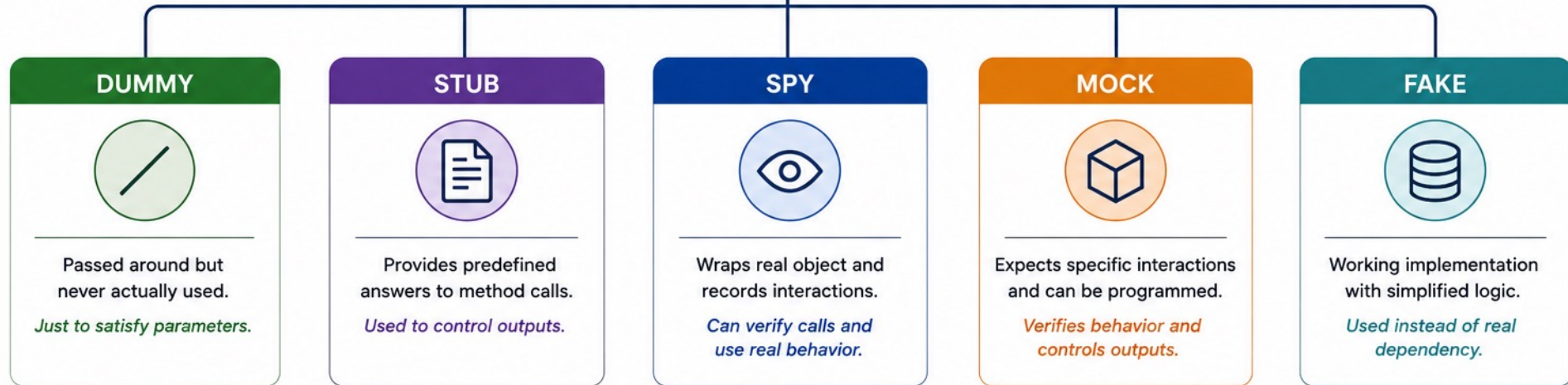
- `mock(List.class)` creates an empty mock — every method returns a default value (0, null, empty) until stubbed.
- `spy(real)` wraps a real instance — unstubbed calls run the real method's actual logic.
- Use `doReturn().when(spy)` to stub a spy; `when(spy.method())` can trigger real behavior first and may throw.
- Prefer a mock when no real logic is needed; use a spy only to override a few methods on real behavior.
- `verify()` works identically on both mocks and spies — interaction tracking doesn't depend on which you use.

**KEY TAKEAWAY** Reach for a mock first — use a spy only when you need most of an object's real behavior with one or two methods overridden.



# TEST DOUBLES

Test doubles are stand-ins for real dependencies used in tests.



## SCOPE OF USE



Filling parameters not used



Returning controlled data / values



Partial mocking and interaction testing



Behavior verification and expectations



Replacing complex or external systems

| TYPE        | DUMMY                     | STUB                           | SPY                                    | MOCK                                        | FAKE                                   |
|-------------|---------------------------|--------------------------------|----------------------------------------|---------------------------------------------|----------------------------------------|
| PURPOSE     | Just to pass objects      | Provide test data              | Observe interactions                   | Verify interactions                         | Work like real but simpler             |
| BEHAVIOR    | No behavior               | Returns programmed values      | Calls real methods (can stub some)     | Does not call real methods (unless stubbed) | Real implementation (simplified)       |
| VERIFIES    | ✗ No                      | ✗ No                           | ✓ Yes                                  | ✓ Yes                                       | ✗ No                                   |
| EXAMPLE USE | Method parameter not used | Repository returning user data | Service method calls and partial mocks | Verify repository.save() was called         | In-memory database, fake email service |

# HOW MOCKITO WORKS

*Mockito's core components work together in a repeatable cycle — from creating mocks to verifying interactions.*



## HOW MOCKITO WORKS — CORRECT STEP ORDER

- 1 Create mocks
- 2 Stub required behavior
- 3 Invoke the class under test
- 4 Assert outcome
- 5 Verify important interactions

*Simple diagram: Test → Real Service → Mock Repository*

Mockito generates runtime test doubles using its mock maker, commonly backed by Byte Buddy. The exact mechanism depends on the mocked type and configuration.

**KEY TAKEAWAY** Every Mockito test follows the same cycle: create mocks, stub required behavior, invoke the class under test, assert the outcome, then verify important interactions — stubbing always comes before invocation.

# MOCKITO WORKFLOW IN CODE

*Stubbing must happen before the method under test runs — reversing the order silently breaks the stub.*

## EXAMPLE

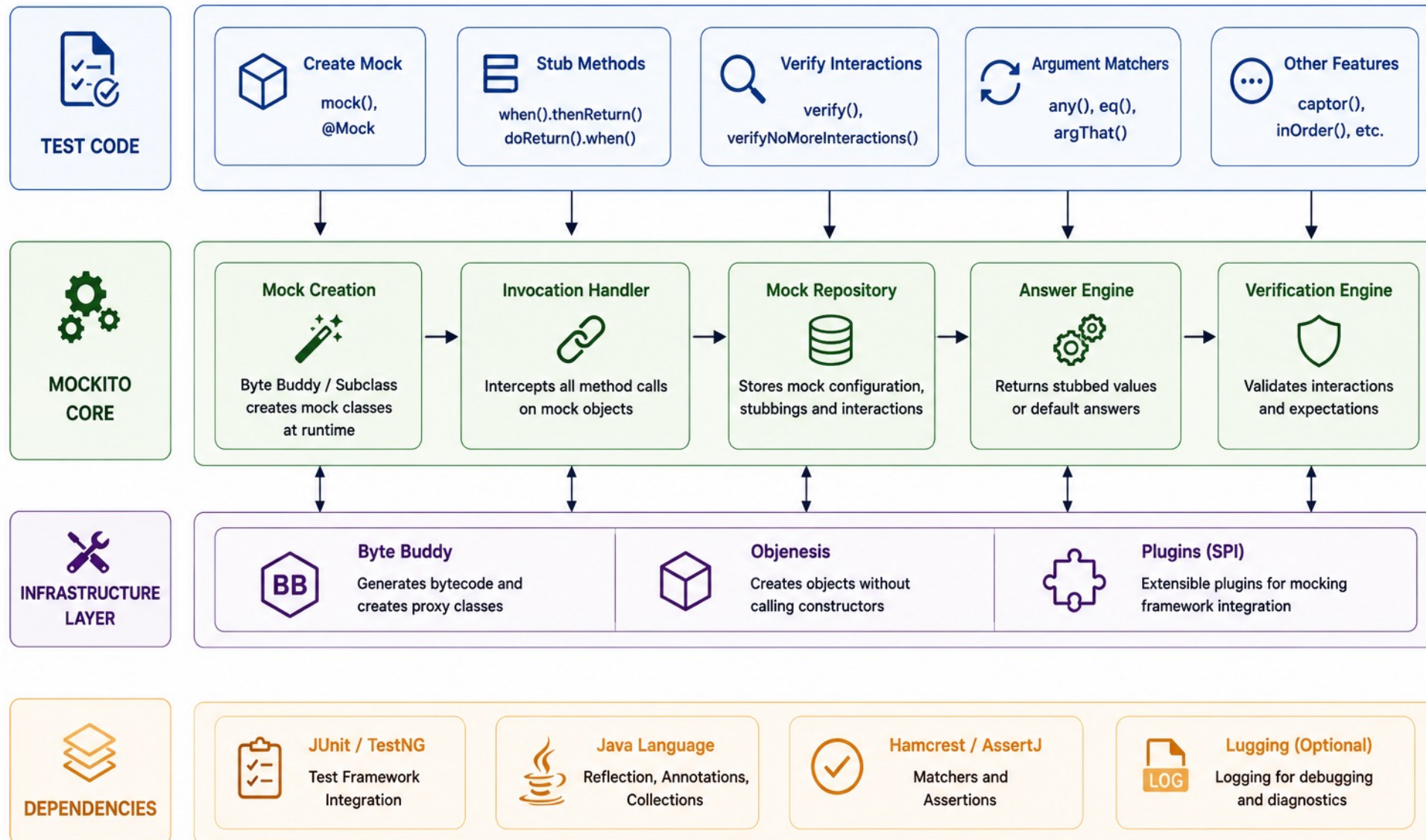
```
// WRONG: stub written after invocation
Customer result = service.getCustomer("CUS-1001");
when(repository.findById("CUS-1001"))
    .thenReturn(Optional.of(customer));
// result is null – the stub came too late

// RIGHT: create, stub, invoke, assert, verify
when(repository.findById("CUS-1001"))
    .thenReturn(Optional.of(customer));
Customer ok = service.getCustomer("CUS-1001");
verify(repository).findById("CUS-1001");
```

**Note:** The mock returns null for any unstubbed call. Calling `getCustomer()` before `when()` means the WRONG block above compiles and passes with a null result — silently.

**KEY TAKEAWAY** Define stubs before you call the method under test — Mockito can't retroactively apply a `when()` to a call that already happened.

# MOCKITO ARCHITECTURE



# CREATING MOCKS

*Mockito provides several ways to create mock objects for your dependencies.*

## EXAMPLE

```
@ExtendWith(MockitoExtension.class)
class DefaultCustomerServiceTest {
    @Mock
    CustomerRepository repository;
    CustomerValidator validator =
        new CustomerValidator();
    DefaultCustomerService service;
    @BeforeEach
    void setUp() {
        service = new DefaultCustomerService(
            repository, validator);
    }
}
```

## KEY POINTS

- @Mock creates a mock object for a class or interface.
- Prefer explicit construction in @BeforeEach — pass mocks and real collaborators into the constructor directly for clarity.
- @InjectMocks is a convenience alternative: it creates the object under test and injects matching mocks automatically.
- @ExtendWith(MockitoExtension.class) is the simplest, primary way to enable annotations in JUnit 5.
- Alternative: Mockito.mock(Class) directly, or MockitoAnnotations.openMocks(this) — if used, close the returned AutoCloseable.
- Advanced: use MockSettings for advanced mock configuration (e.g., extra interfaces, custom answers).
- Convenience snippet: @InjectMocks DefaultCustomerService service; — declared once, wired automatically once all constructor dependencies are mocked or set.

**KEY TAKEAWAY** Mock dependencies whose real behavior would make the test slow, nondeterministic, difficult to control, or outside the unit's responsibility. Use real objects for simple, deterministic collaborators where that improves clarity.

# @INJECTMOCKS & MOCK()

*Two more ways to create mocks: automatic field injection, and a mock built without any annotation.*

## EXAMPLE

```
@ExtendWith(MockitoExtension.class)
class DefaultCustomerServiceTest {
    @Mock
    CustomerRepository repository;
    @Mock
    CustomerNotifier notifier;

    @InjectMocks
    DefaultCustomerService service;
}

CustomerRepository repo =
    mock(CustomerRepository.class);
```

## KEY POINTS

- @InjectMocks creates DefaultCustomerService and injects the @Mock fields above into its constructor automatically.
- Field injection only works when Mockito can match mocks to constructor parameters unambiguously.
- Mockito.mock(Class) creates a mock directly, with no annotation or extension — useful outside test classes.
- Prefer explicit constructor wiring (shown earlier) for clarity; use @InjectMocks for classes with many collaborators.
- Both approaches produce the same kind of mock — the difference is only how it's created and wired.

**KEY TAKEAWAY** @InjectMocks and Mockito.mock() are conveniences, not requirements — explicit constructor wiring stays the clearest choice for a few dependencies.

# TRY IT YOURSELF — EXERCISE 1: WHEN TO KEEP REAL VALIDATOR

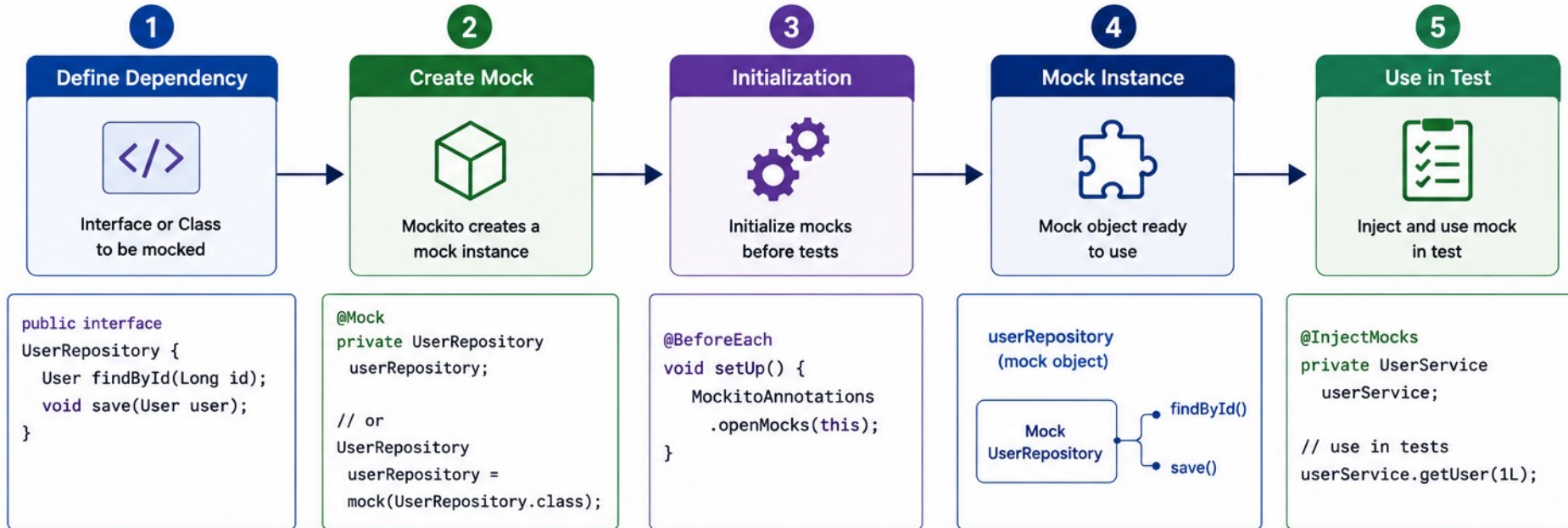
*Reinforces: deciding which collaborators to mock vs. keep real — like the CustomerValidator you just saw wired in directly.*

| STEP        | WHAT TO DO                                                                                          |
|-------------|-----------------------------------------------------------------------------------------------------|
| Goal        | Decide which activate() collaborator stays real and which gets mocked                               |
| File        | mock-real-decisions.md (in examples/module-18-exercises/notes/)                                     |
| Mock        | CustomerRepository — an I/O boundary; slow, non-deterministic, outside the unit                     |
| Keep real   | StatusValidator stays real when it is deterministic, fast, and pure domain logic                    |
| Deliverable | Repo-mock justified, validator-real justified, notifier-mock justified — a three-row decision table |
| Reference   | exercises/exercise-01-keep-real-validator.md                                                        |

```
$ cat mock-real-decisions.md
CustomerRepository -> MOCK    // I/O boundary, slow / non-deterministic
StatusValidator   -> REAL    // pure, deterministic, fast
CustomerNotifier  -> MOCK    // avoid email/IO in unit tests
RULE: mock I/O and unstable deps; keep pure domain helpers real when cheap
```

**TRY IT NOW** Complete Exercise 1 before continuing — see exercises/exercise-01-keep-real-validator.md.

# CREATING MOCKS



## WAYS TO CREATE MOCKS



## RESULT



# STUBBING METHODS

*Stubbing defines what a mock should return (or throw) when a method is called.*

## EXAMPLE

```
when(orderRepository.findById(1L))
    .thenReturn(Optional.of(order));
when(paymentGateway.charge(any()))
    .thenThrow(new PaymentException());

doReturn(value).when(spy).method();
doThrow(new NotificationException())
    .when(notificationService).send(any());

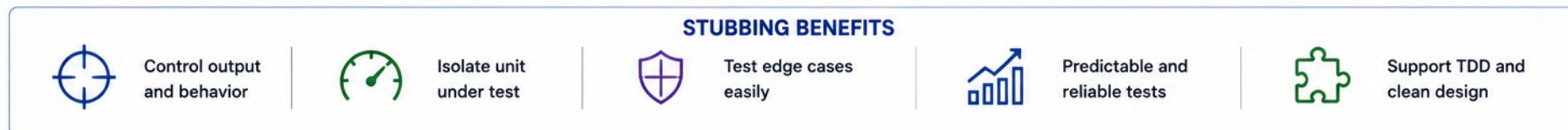
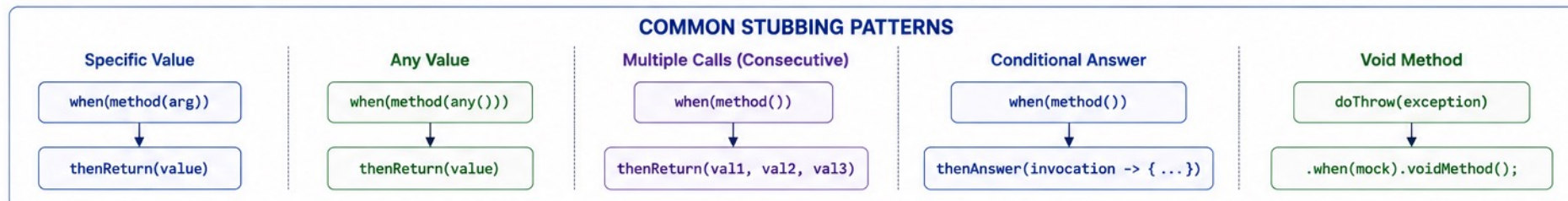
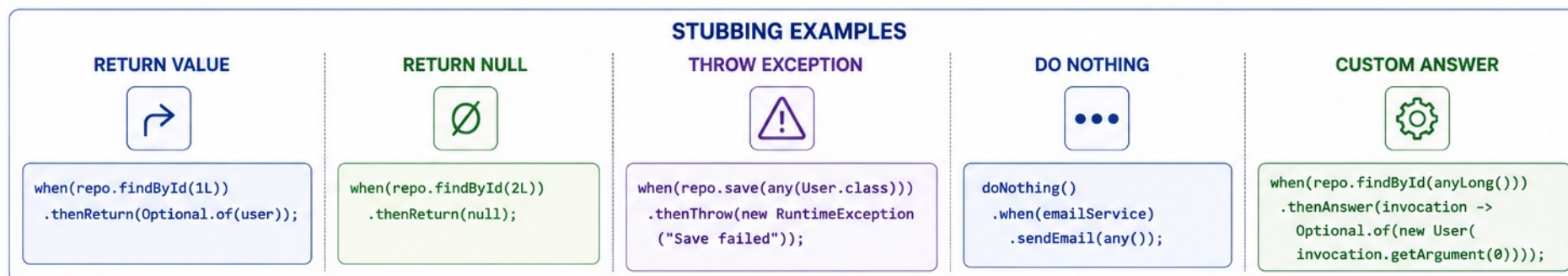
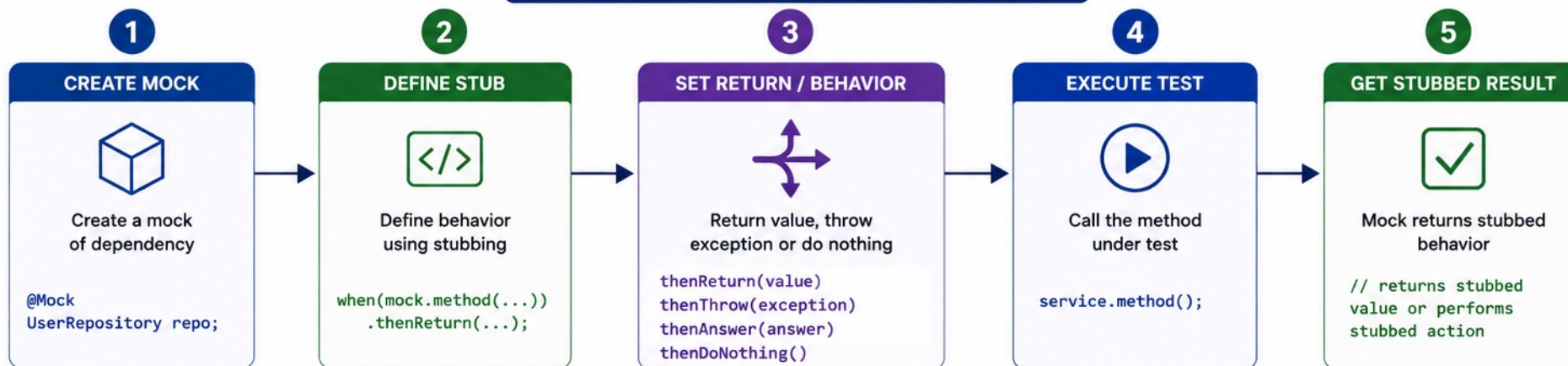
when(repository.findByStatusAndRegion(
    eq(ACTIVE), anyString()))
    .thenReturn(customers);
```

## COMMON STUBBING METHODS

- `when(mock.method()).thenReturn(value)` — normal flow.
- `when(mock.method()).thenThrow(exception)` — exceptional flow.
- For void methods, use `doNothing()`, `doThrow()`, or `doAnswer()` — never `doReturn()`.
- `doReturn()` is commonly useful for spies, or when `when()` would invoke real behavior.
- Use `doNothing()` only when it improves readability or is part of consecutive behavior. For ordinary mock void methods, no stubbing is required.
- Stub only what is necessary for the test.
- Advanced: `thenReturn(v1, v2, v3)` returns different values per call — can couple a test to invocation order.
- When one argument uses a matcher, all arguments in that invocation should normally use matchers.

**KEY TAKEAWAY** Stub only the behavior your test actually needs — over-stubbing makes tests brittle and hard to maintain.

# STUBBING METHODS



# VERIFYING INTERACTIONS

*Verification confirms that the unit under test called its dependencies as expected.*

| METHOD                                      | PURPOSE                                    | EXAMPLE                                                                     |
|---------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------|
| <code>verify(mock).method(args)</code>      | Confirms method was called once with args  | <code>verify(repo).save(user)</code>                                        |
| <code>verify(mock, times(n))</code>         | Confirms method was called exactly n times | <code>verify(repo, times(2)).save(any())</code>                             |
| <code>verify(mock, never())</code>          | Confirms method was never called           | <code>verify(gateway, never()).charge(any())</code>                         |
| ADVANCED VERIFICATION                       |                                            |                                                                             |
| <code>verify(mock, atLeastOnce())</code>    | Confirms method was called at least once   | <code>verify(notificationService).sendCustomerActivated(customerId);</code> |
| <code>inOrder(...).verify(...)</code>       | Confirms methods were called in sequence   | <code>inOrder.verify(repo).findById(1L)</code>                              |
| <code>verifyNoMoreInteractions(mock)</code> | Confirms no other interactions occurred    | <code>verifyNoMoreInteractions(userRepository)</code>                       |

```
ArgumentCaptor<Customer> captor =
    ArgumentCaptor.forClass(Customer.class);
verify(repository).save(captor.capture());
Customer saved = captor.getValue();
assertEquals(ACTIVE, saved.status());
```

**Captor:** Use a captor when the argument itself is an important observable outcome. Do not use it when a direct argument equality check is enough.

**Results first:** Assert the observable result first. Verify interactions only when they are part of the expected behavior. Often: result, exception, entity state, captured object, side effect.

**atLeastOnce():** Prefer exact verification when the call count is part of the behavior. Use `atLeastOnce()` only when additional calls are intentionally acceptable.

**verifyNoMoreInteractions():** Use it only when interaction completeness is part of the contract, such as security-sensitive side effects or ensuring no persistence occurs after rejection.

**KEY TAKEAWAY** Verify behavior, not implementation — confirm the important interactions happened, not every internal detail.

# TRY IT YOURSELF — EXERCISE 2: STUB VS VERIFY

*Reinforces: stubbing return values with when() vs. verifying side-effect calls with verify() — both covered above.*

| STEP        | WHAT TO DO                                                                   |
|-------------|------------------------------------------------------------------------------|
| Goal        | Explain stubbing return values vs. verifying calls for activate()            |
| File        | lab18-stub-verify.md (in examples/module-18-exercises/notes/)                |
| Stub        | when(repo.findById("CUS-1002")).thenReturn(raviProspect) — arrange the input |
| Verify      | verify(repo).save(activatedRavi) — assert the collaboration happened         |
| Deliverable | A written stub example, a written verify example, and one contrast sentence  |
| Reference   | exercises/exercise-02-stub-vs-verify.md                                      |

```
$ cat lab18-stub-verify.md
when(repo.findById("CUS-1002"))    // STUB: arrange an input
    .thenReturn(raviProspect);
verify(repo).save(activatedRavi);  // VERIFY: assert a side effect
// Stubs feed inputs; verifies prove side-effect calls happened.
```

**TRY IT NOW** Complete Exercise 2 before continuing — see exercises/exercise-02-stub-vs-verify.md.

# TRY IT YOURSELF — EXERCISE 3: ARGUMENTCAPTOR PREVIEW

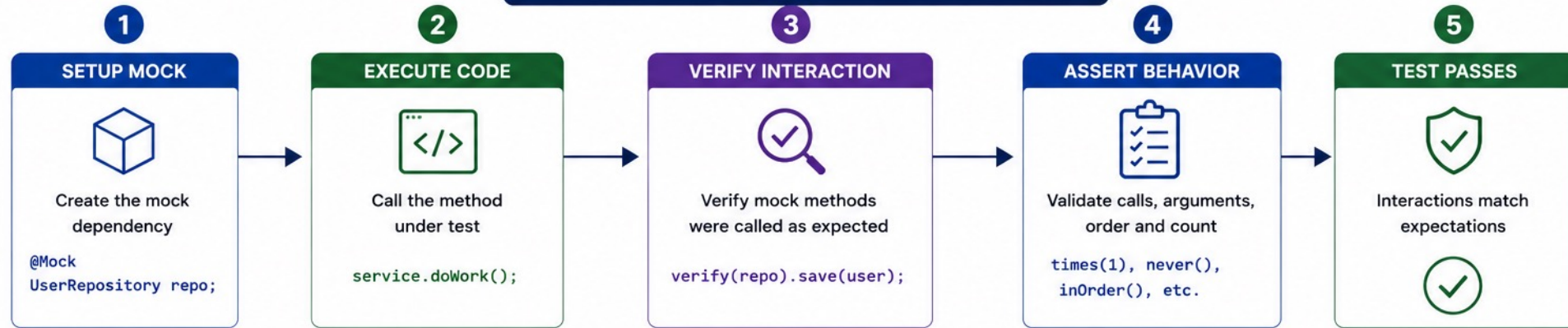
*Reinforces: the ArgumentCaptor pattern shown above for inspecting a saved Customer's state, not just that save() ran.*

| STEP            | WHAT TO DO                                                                                |
|-----------------|-------------------------------------------------------------------------------------------|
| Goal            | Sketch ArgumentCaptor steps for the saved Customer, on paper — no tests run yet           |
| File            | lab18-captor-sketch.md (in examples/module-18-exercises/notes/)                           |
| Declare         | ArgumentCaptor<Customer> captor = ArgumentCaptor.forClass(Customer.class);                |
| Verify + assert | verify(repo).save(captor.capture()); then assert captor.getValue().getStatus() is ACTIVE  |
| Deliverable     | A three-step captor sketch, ACTIVE asserted for Ravi, and a written pre-lab boundary note |
| Reference       | exercises/exercise-03-argumentcaptor-preview.md                                           |

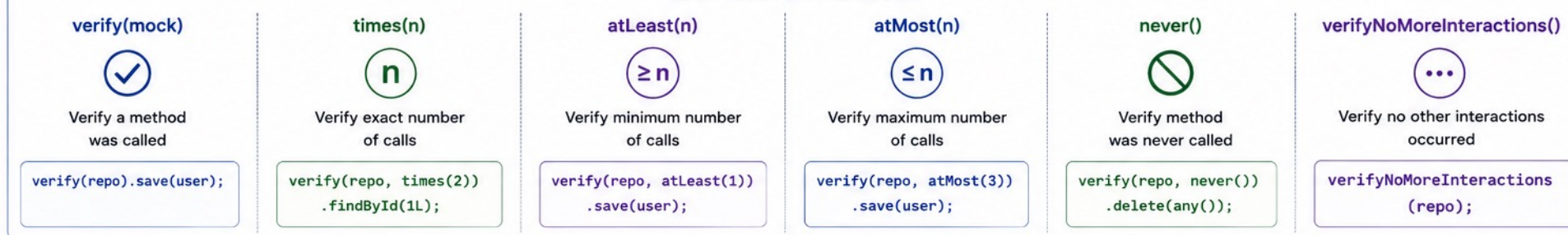
```
$ cat lab18-captor-sketch.md
ArgumentCaptor<Customer> captor = ArgumentCaptor.forClass(Customer.class);
verify(repository).save(captor.capture());
assertEquals(ACTIVE, captor.getValue().getStatus());
// Prepare for Lab 18; do not complete the full Mockito lab now.
```

**TRY IT NOW** Complete Exercise 3 before continuing — see exercises/exercise-03-argumentcaptor-preview.md.

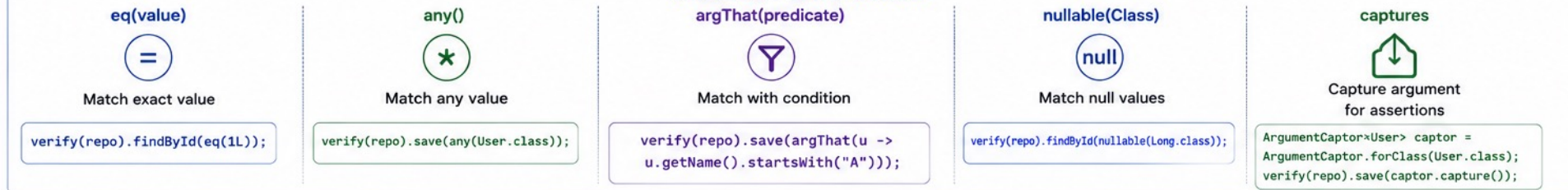
# VERIFYING INTERACTIONS



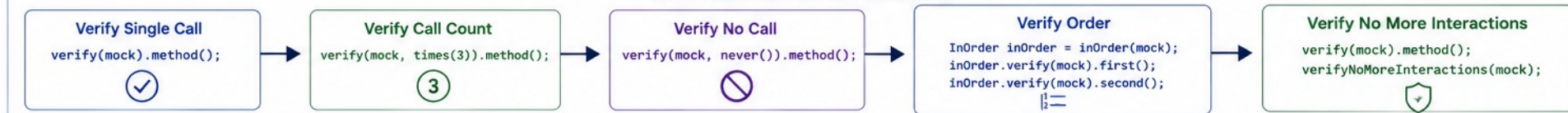
## VERIFICATION METHODS



## ARGUMENT VERIFICATION



## COMMON VERIFICATION PATTERNS



# MOCKING REPOSITORIES

*Repositories abstract data access — mock them to control responses and isolate business logic from the database.*

## EXAMPLE

```
@Test
void getCustomer_whenCustomerExists_returnsCustomer() {
    when(repository.findById("CUS-1001"))
        .thenReturn(Optional.of(customer));

    Customer result =
        service.getCustomer("CUS-1001");

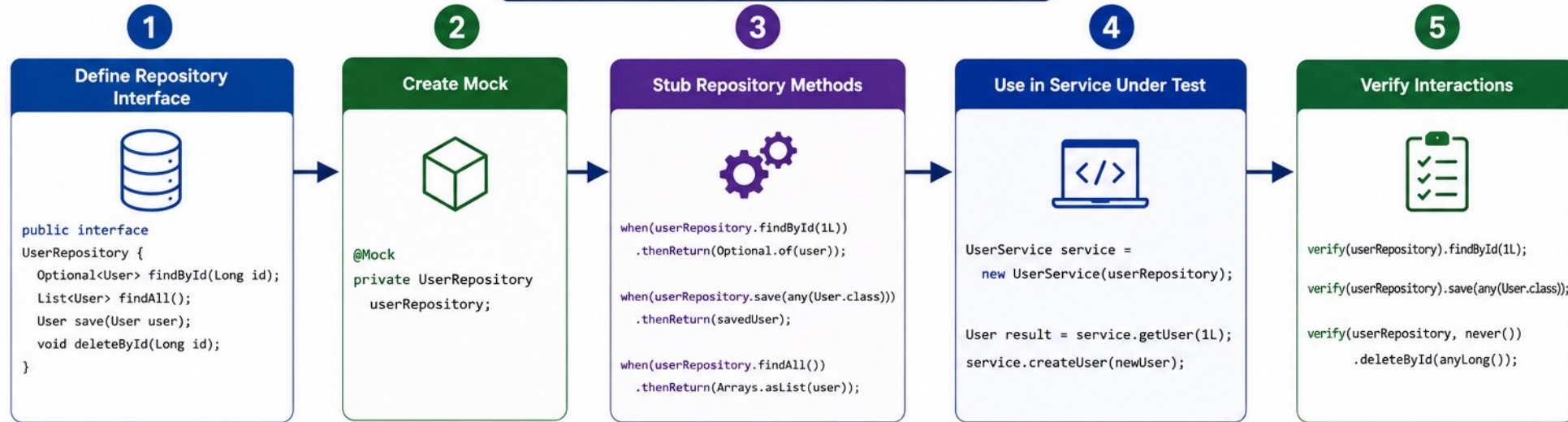
    assertAll(
        () -> assertEquals("CUS-1001", result.id()),
        () -> assertEquals("Amina Khan", result.name())
    );

    verify(repository).findById("CUS-1001");
}
```

**Note:** Mockito verifies service behavior, not persistence integration. Repository behavior still requires integration tests against the real persistence technology.

**KEY TAKEAWAY** Prefer real domain objects and value objects. Mock infrastructure boundaries and collaborators outside the behavior being tested.

# MOCKING REPOSITORIES



## COMMON REPOSITORY STUBBING

### Find By ID



```
when(repo.findById(1L))
    .thenReturn(Optional.of(user));

when(repo.findById(99L))
    .thenReturn(Optional.empty());
```

### Find All



```
when(repo.findAll())
    .thenReturn(Arrays.asList(
        user1, user2
    ));
```

### Save



```
when(repo.save(any(User.class)))
    .thenAnswer(invocation -> {
        User u = invocation.getArgument(0);
        u.setId(1L);
        return u;
    });
```

### Delete



```
doNothing().when(repo)
    .deleteById(anyLong());

doThrow(new RuntimeException("DB Error"))
    .when(repo).deleteById(anyLong());
```

### existsBy... / Count



```
when(repo.existsById(1L))
    .thenReturn(true);

when(repo.count())
    .thenReturn(5L);
```

## BENEFITS OF MOCKING REPOSITORIES



Isolates business logic from database



Faster and reliable unit tests



Control data and scenarios easily



Test edge cases and error paths



Verify interactions with persistence layer

# MOCKING EXTERNAL SERVICES

*Mock external service calls (payment gateways, email/notifications, third-party APIs) to test without real network calls.*



## Payment Gateways

- Stripe, PayPal, Braintree. Mock the gateway interface, not the SDK — test approved and declined responses.
- `interface PaymentGateway {`
- `PaymentResult charge(...); }`



## Notification Services

- Email, SMS, push. Wrap the vendor client in an app-owned port; mock the port, not the SDK.
- `interface CustomerNotificationGateway {`
- `void sendActivated(Customer c); }`



## Third-Party APIs

- Partner and internal REST APIs. Mock your own gateway interface, not deep framework chains (e.g., WebClient) — simulate timeouts and 4xx/5xx errors at that boundary.

## BEST PRACTICE

- Return success and failure responses to test both the happy path and error-handling logic in your service.
- Blind spot: Use Mockito for application behavior. Use HTTP stubs, contract tests, or integration tests for protocol and payload correctness.

**KEY TAKEAWAY** Mock the application-owned boundary interface, not the vendor SDK or framework client — keep tests focused on your own logic, not third-party protocols.

# STUBBING A PAYMENT GATEWAY

*Mock the boundary interface and stub both the approved and declined payment paths.*

## EXAMPLE

```
@Mock
PaymentGateway paymentGateway;

@Test
void charge_whenDeclined_throwsException() {
    when(paymentGateway.charge(any()))
        .thenThrow(new PaymentException(
            "Card declined"));

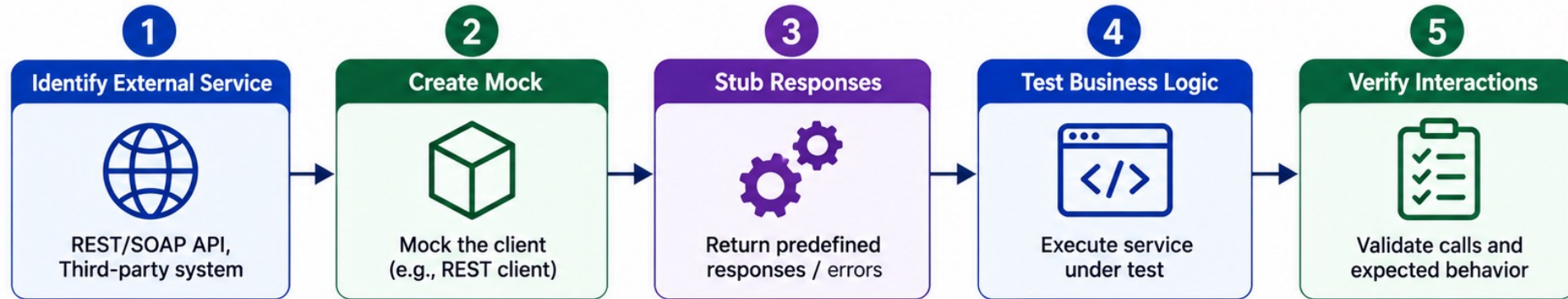
    assertThrows(PaymentException.class,
        () -> checkout.pay(order));
    verify(paymentGateway).charge(any());
}
```

## KEY POINTS

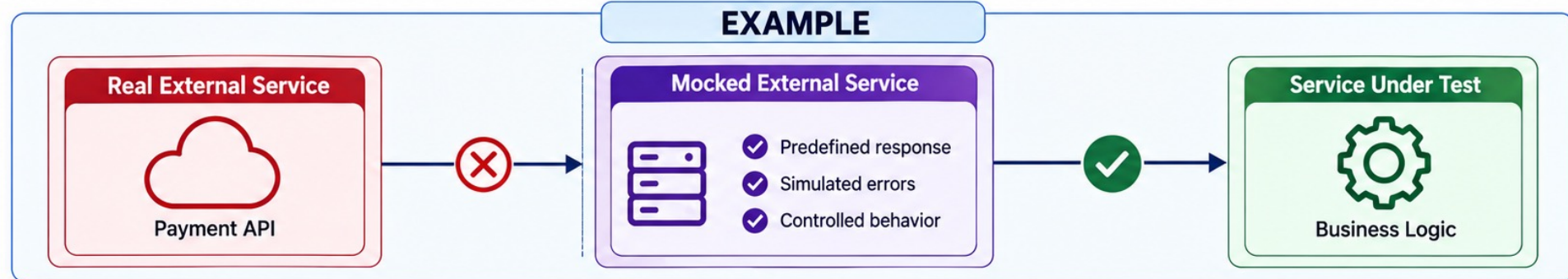
- Mock the application-owned PaymentGateway interface — never the Stripe/PayPal SDK client directly.
- thenThrow() simulates a declined payment or network failure without making a real call.
- assertThrows() confirms the service translates the gateway exception into the expected behavior.
- any() matches any argument — use it when the exact charge request isn't the focus of this test.
- verify() confirms the gateway was actually invoked, not just that an exception was thrown some other way.

**KEY TAKEAWAY** Test both outcomes of an external boundary — thenReturn() for the happy path, thenThrow() for failures — without a single real network call.

# MOCKING EXTERNAL SERVICES



## EXAMPLE



## BENEFITS



# MOCKING SERVICE LAYERS

*Mock a service dependency when testing a controller or another service that calls it.*

## EXAMPLE

```
@Mock
private PricingService pricingService;

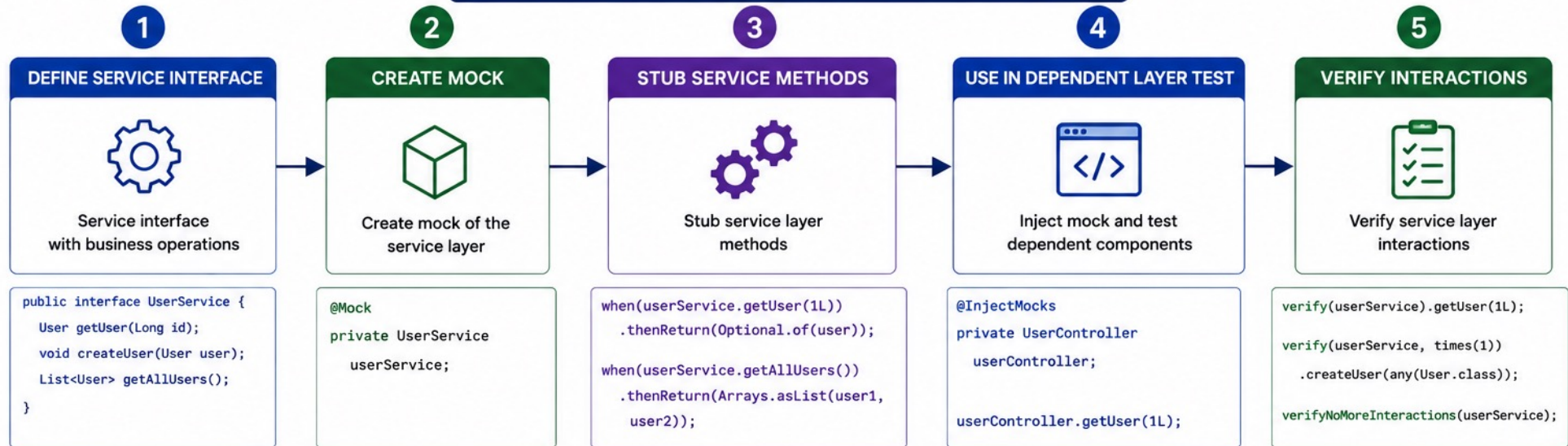
when(pricingService.calculateTotal(order))
    .thenReturn(new BigDecimal("199.99"));
```

## WHY MOCK A SERVICE LAYER

- Test one layer at a time — the controller, not the service it calls.
- Avoid cascading real logic across multiple layers in a single test.
- Makes it easy to simulate edge cases the dependent service would produce.
- Mock an application service when testing a controller adapter. When testing one service that collaborates with another, consider whether the second collaborator is truly outside the unit or whether a broader component test would provide better confidence.

**KEY TAKEAWAY** Mocking service layers keeps each test focused on a single unit, even in a multi-layered architecture.

# MOCKING SERVICE LAYERS



## COMMON STUBBING PATTERNS

### RETURN VALUE



```
when(service.method())
    .thenReturn(value);
```

### RETURN NULL



```
when(service.method())
    .thenReturn(null);
```

### THROW EXCEPTION



```
when(service.method())
    .thenThrow(new
    RuntimeException("Error"));
```

### RETURN COLLECTION



```
when(service.getAll())
    .thenReturn(Arrays.asList(
    item1, item2));
```

### CONDITIONAL STUBBING



```
when(service.calculate(anyInt()))
    .thenAnswer(invocation -> {
        int arg = invocation.getArgument(0);
        return arg * 2;
    });
```

## BENEFITS



Isolates business logic



Faster and reliable tests



No dependency on complex services



Control inputs and scenarios



Easier testing of edge cases

# ISOLATED SERVICE TEST WORKFLOW

*A repeatable seven-step workflow for testing business logic against real domain data and mocked boundaries.*

- 1 Arrange real domain data
- 2 Mock external boundaries
- 3 Stub only required behavior
- 4 Call the real service
- 5 Assert returned state or exception
- 6 Verify meaningful interactions
- 7 Confirm forbidden interactions did not occur

## WHAT TO TEST

- Success paths (order placed, payment succeeds) and failure paths (payment declined, item not found) — both matter equally.

## BEST PRACTICES

- Follow the AAA pattern: Arrange, Act, Assert.
- Test one business scenario per test.
- Use meaningful test data.
- Keep tests small, focused, and readable.

**KEY TAKEAWAY** Follow the same seven-step workflow for every isolated service test — from arranging real data to confirming forbidden interactions never happened.

# ISOLATED WORKFLOW IN CODE

*The same seven steps as one Mockito test — from arranging real data to confirming forbidden interactions.*

## EXAMPLE

```
@Test
void activate_whenProspect_activatesAndNotifies() {
    Customer ravi =
        Customer.prospect("CUS-1002", "Ravi Shah");

    when(repository.findById("CUS-1002"))
        .thenReturn(Optional.of(ravi));

    Customer activated = service.activate("CUS-1002");
    assertEquals(ACTIVE, activated.status());
    verify(repository).save(activated);
    verify(notifier).sendActivated(activated);

    verify(paymentGateway, never()).charge(any());
}
```

**KEY TAKEAWAY** Follow the same seven-step workflow for every isolated service test — arrange, mock, stub, call, assert, verify, then confirm forbidden interactions never happened.

# TRY IT YOURSELF — EXERCISE 4: ACTIVATE TODOs (STARTER)

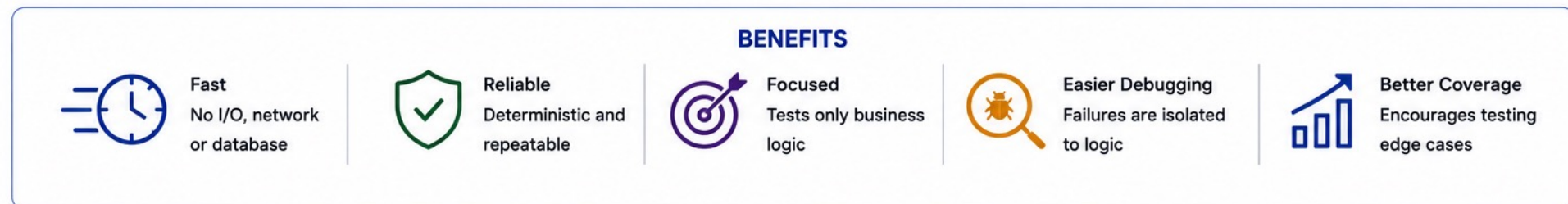
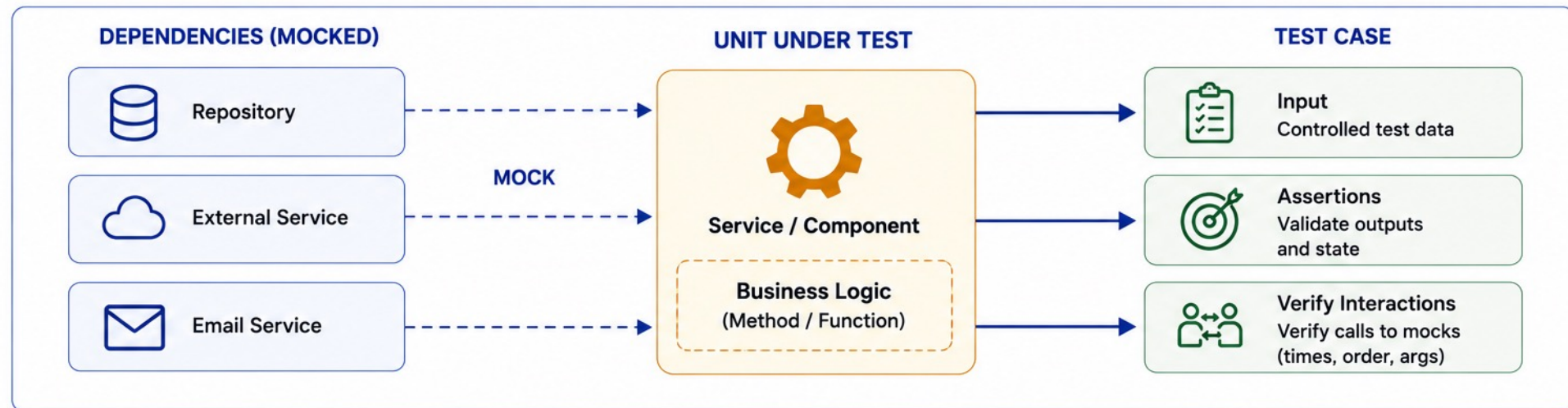
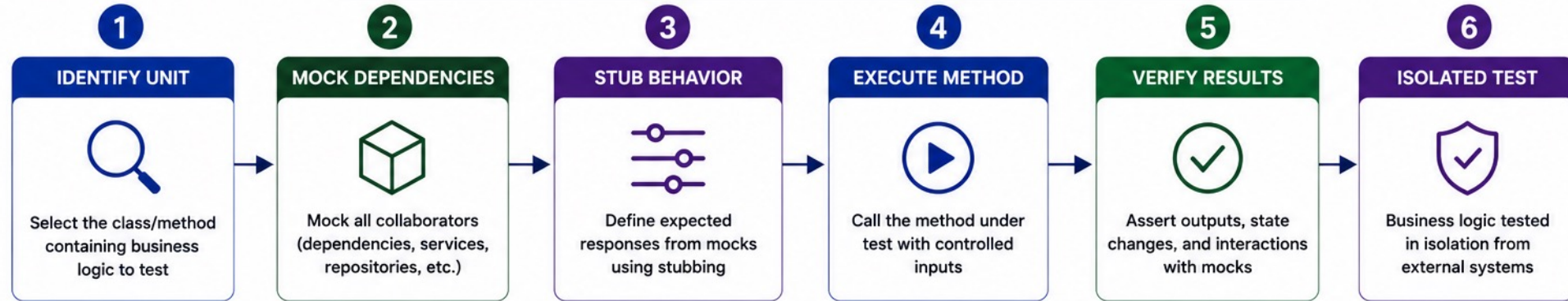
*Reinforces: the seven-step isolated service test workflow you just saw, applied to the activate() interaction sequence.*

| STEP        | WHAT TO DO                                                                                               |
|-------------|----------------------------------------------------------------------------------------------------------|
| Goal        | FILL-IN-THE-BLANK STARTER — complete six blanks in the activate() interaction sequence                   |
| File        | lab18-interaction-todos.md (in examples/module-18-exercises/notes/)                                      |
| Fill in     | Six blanks: the customer id, the method name, two verify() calls, the final status, and the captor field |
| Self-check  | Step 1's id must read CUS-1002; the final status in steps 5 and 6 must read ACTIVE                       |
| Deliverable | All six blanks replaced correctly, plus one sentence on the captor's benefit                             |
| Reference   | exercises/exercise-04-fill-activate-interaction-todos.md                                                 |

```
1) stub findById(____) -> ravi PROSPECT
2) call service.____(...)
3) verify repo.____(customer)  4) verify notifier.____(...)  // if present
5) assert status ____
6) captor previews status field ____
```

**TRY IT NOW** Complete Exercise 4 before continuing (fill in all six blanks) — see exercises/exercise-04-fill-activate-interaction-todos.md.

# TESTING BUSINESS LOGIC IN ISOLATION



# COMMON MOCKING MISTAKES

*Avoid these pitfalls to keep your mocked tests reliable and meaningful.*

| MISTAKE                                        | PROBLEM                                          | BETTER APPROACH                                        |
|------------------------------------------------|--------------------------------------------------|--------------------------------------------------------|
| Over-mocking                                   | Tests verify implementation, not behavior        | Mock only true external dependencies                   |
| Over-stubbing                                  | Stubs unused by the test add noise and confusion | Stub only what the test actually needs                 |
| Verifying too much                             | Brittle tests break on harmless refactors        | Verify only interactions that matter                   |
| Mocking value objects / DTOs                   | Unnecessary and misleading                       | Use real instances for simple data objects             |
| Ignoring <code>verifyNoMoreInteractions</code> | Missed unexpected calls go unnoticed             | Use it when interaction completeness matters           |
| Mocking the class under test                   | No real behavior is exercised                    | Instantiate the real class and mock only collaborators |

**KEY TAKEAWAY** Mock dependencies, not the logic you're trying to test — mocking too much hides real bugs.

# MISTAKE VS. FIX IN CODE

*Verifying every call locks a test to today's implementation — verify only what the test's behavior depends on.*

## EXAMPLE

```
// MISTAKE: verifies every internal call
verify(repository).findById("CUS-1002");
verify validator.validate(any());
verify(repository).save(any());
verify(notifier).sendActivated(any());
verify(auditLog).record(any());

// BETTER: verify only what the
// test's behavior depends on
verify(repository).save(activated);
verify(notifier).sendActivated(activated);
```

## KEY POINTS

- The mistake block treats “verify everything” as a safety net — it actually locks the test to today's implementation.
- A harmless refactor (e.g., adding an audit log call) now breaks a test that never cared about auditing.
- The fix verifies only the two interactions the test's name and assertion actually depend on.
- This is the “Verifying too much” row from Common Mocking Mistakes: brittle tests break on harmless refactors.
- Rule of thumb: if deleting a verify() wouldn't let a wrong implementation pass, it isn't testing real behavior.

**KEY TAKEAWAY** Every verify() should protect a real behavior — if you can delete it without weakening the test, delete it.

# TRY IT YOURSELF — EXERCISE 5: MOCKITO ANTI-PATTERNS

*Reinforces: the common mocking mistakes table you just saw — now applied to Copilot-suggested Mockito code.*

| STEP           | WHAT TO DO                                                                                          |
|----------------|-----------------------------------------------------------------------------------------------------|
| Goal           | List anti-patterns an AI assistant might suggest for CRM tests, and how to reject them              |
| File           | lab18-anti-patterns.md (in examples/module-18-exercises/notes/)                                     |
| Table          | Recreate the mistake/better table and add one row: mocking a String or enum status is silly         |
| AI reject rule | Reject any suggestion that mocks CustomerService while the test is testing CustomerService          |
| Deliverable    | The extended table, the SUT-mock reject rule, and a written preference for real Amina/Ravi fixtures |
| Reference      | exercises/exercise-05-anti-patterns.md                                                              |

```
$ cat lab18-anti-patterns.md
Mock the SUT                -> Mock collaborators only
Unnecessary stubbing         -> Stub only what the test uses
verifyNoMoreInteractions always -> Use only when the interaction surface is critical
Mocking String/enum status   -> Silly – just use the real value
```

**TRY IT NOW** Complete Exercise 5 before continuing — see exercises/exercise-05-anti-patterns.md.

# TRY IT YOURSELF — EXERCISE 6: LAB 18 PREP CHECKLIST

*Reinforces: everything from Exercises 1–5 — confirm you're ready before Lab 18 begins.*

| STEP        | WHAT TO DO                                                                                      |
|-------------|-------------------------------------------------------------------------------------------------|
| Goal        | Confirm Mockito pre-lab prep is complete without starting Lab 18 itself                         |
| File        | lab18-prep-checklist.md (in examples/module-18-exercises/notes/)                                |
| Confirm     | Stub/verify note, mock/real decision table, anti-pattern sheet, and interaction TODOs all exist |
| Boundary    | Write: "Pre-lab only — prepare for lab; do not complete full Lab 18."                           |
| Deliverable | A completed readiness checklist, captor preview confirmed, and the Lab 19 pointer noted         |
| Reference   | exercises/exercise-06-lab18-prep-checklist.md                                                   |

```
[x] Exercise 2 - stub vs verify note saved
[x] Exercise 1 - mock/real decision table saved
[x] Exercise 5 - anti-pattern sheet saved
[x] Exercise 4 - interaction TODOs filled
[x] Exercise 3 - captor sketch saved
```

**TRY IT NOW** Complete Exercise 6 before Lab 18 — see exercises/exercise-06-lab18-prep-checklist.md.

# LAB OVERVIEW – MOCKITO AND MOCKING WITH AI ASSISTANCE

*Use Mockito to create mocks, stub methods, verify interactions, and test business logic in isolation.*

## SCENARIO

- Isolate DefaultCustomerService (Northstar CRM) by mocking CustomerRepository — keep CustomerValidator real so uniqueness rules still run.
- Verify find/save interactions, capture saved entities with ArgumentCaptor, and prove not-found paths never call save.

## TECH STACK

| ITEM     | VALUE            |
|----------|------------------|
| Language | Java 21          |
| Testing  | JUnit 5, Mockito |
| Build    | Maven            |
| IDE      | IntelliJ IDEA    |

**KEY TAKEAWAY** By the end of this lab, you'll write effective unit tests using Mockito and apply mocking techniques in real projects.

# LAB TASKS AND DELIVERABLES

Complete these tasks to gain hands-on experience with Mockito — each builds on the previous one.

| # | TASK                 | KEY ACTIVITIES                                                                                                                                                                  | FOCUS AREA               |
|---|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1 | Create Mocks         | Use @Mock for each dependency; initialize with @ExtendWith(MockitoExtension.class).                                                                                             | Mock setup               |
| 2 | Stub Methods         | Use when().thenReturn()/thenThrow(); stub only what's necessary.                                                                                                                | Behavior definition      |
| 3 | Verify Interactions  | Use verify(mock).method(), times(), never(), atLeastOnce().                                                                                                                     | Interaction verification |
| 4 | Mock Repositories    | Stub scenario-by-scenario: existing lookup, missing lookup, save — only what the service actually calls.                                                                        | Data access isolation    |
| 5 | ArgumentCaptor & BDD | Capture the saved Customer; assert ID, name/email, status, and a single save. Add CustomerServiceBddMockTest using given()/then().should().                                     | Captor + BDD style       |
| 6 | Prove Isolation      | verify(repository, never()).save(any(Customer.class)) and the thrown exception for CUS-9999 (or verifyNoInteractions(repository)). Run individually, as a suite, and reordered. | Negative paths & docs    |

### SUCCESS CRITERIA

- CustomerServiceMockitoTest and CustomerServiceBddMockTest with a mocked CustomerRepository; verified interactions, ArgumentCaptor, and never().save() on not-found; clean, readable test code.

### AI ASSISTANCE (OPTIONAL)

- Use GitHub Copilot to draft a mock setup, then review it: reject drafts that mock the class under test, leave unused stubs, or add Thread.sleep. Log: the prompt used, whether the suggestion was accepted or rejected and why, and any hallucinated API or unnecessary stub identified.

**KEY TAKEAWAY** Apply Mockito to create clean, isolated, reliable unit tests that validate behavior, not implementation.

# MOCKITO TEST: DEFINITION OF DONE

*Use this checklist to confirm an isolated Mockito test is complete before it ships.*

- 1 Real class under test is instantiated
- 2 Unit boundary is clear
- 3 Simple domain objects remain real
- 4 Only external or controlled collaborators are replaced
- 5 Stubs are minimal and scenario-specific
- 6 Observable result is asserted
- 7 Important interactions are verified
- 8 Forbidden side effects are verified where relevant
- 9 Captors are used only when arguments matter
- 10 No unused stubs remain
- 11 Test does not depend on invocation order unless order is contractual
- 12 Repository and network integration are tested elsewhere
- 13 AI-generated Mockito code has been reviewed

**KEY TAKEAWAY** Testing in isolation leads to faster, more reliable, and maintainable tests that focus on behavior, not implementation.

# KNOWLEDGE CHECK (1 OF 2)

*Test your understanding of Mockito and test isolation.*

**1. A test mocks Customer, CustomerValidator, CustomerRepository, and DefaultCustomerService itself. What is the primary problem?**

- A. Missing @ExtendWith(MockitoExtension.class)
- B. The mocked class means no real logic runs.**
- C. Too many dependencies are mocked at once
- D. CustomerValidator should not be mocked

**3. What does @InjectMocks do?**

- A. Creates a mock for the class under test
- B. Injects created mocks into the real object under test**
- C. Verifies method calls
- D. Starts a database connection

**2. A service returns the correct result, but the test verifies six internal repository calls not part of its public behavior. What risk does this create?**

- A. The test will fail intermittently
- B. A brittle, implementation-coupled test.**
- C. The test runs slower than necessary
- D. Mockito throws an UnnecessaryStubbingException

**4. Best way to verify a method was never called?**

- A. verify(mock, never()).method()**
- B. verify(mock, times(0))
- C. assertNull(mock)
- D. when(mock).never()

**KEY TAKEAWAY** Mockito's @Mock, when()/thenReturn(), and verify() form the core toolkit for isolated unit testing.

# KNOWLEDGE CHECK (2 OF 2)

*Four more questions, plus the full answer key.*

## 5. What is a common mocking mistake?

- A. Stubbing only what's needed
- B. Mocking every dependency including value objects**
- C. Verifying important interactions
- D. Using meaningful test names

## 7. What should you verify in a well-designed test?

- A. Every internal method call
- B. Only interactions that matter to the test's purpose**
- C. Private field values
- D. Nothing — verification is optional

### ANSWER KEY

- 1-B 2-B 3-B 4-A 5-B 6-B 7-B 8-B

## 6. Why mock external services like payment gateways?

- A. To slow down the test suite
- B. To avoid real network calls and simulate failures easily**
- C. To make tests depend on live systems
- D. To replace unit tests entirely

## 8. What is the main benefit of testing in isolation?

- A. Tests run against production data
- B. Fast, deterministic tests that pinpoint real bugs**
- C. Tests require a live database
- D. Tests take longer but are more thorough

**KEY TAKEAWAY** Mockito enables test isolation by mocking dependencies — the result is faster, more reliable, more maintainable tests.

## MODULE 18 COMPLETE

# Mockito for Test Isolation

You can now use Mockito confidently to write clean, isolated unit tests that validate behavior, not implementation.